BARNES finger joints

Analysis summary, design rationale and print decision — all results to date, on one material basis (SUNLU TPU 90A)
Branch claude/modest-cray-kbb75b · generated by scripts/build_print_report.py

1. What to print, in one page

This is the list of parts worth printing next, drawn from every analysis in the repository: the A3, A6 and A8 Type A springs, the 55 round-1 Type B variants, the tendon-routing studies, and the round-2 tendon-load FEA and A8 mesh-convergence check run for this report. Several designs have more than one variant on the list where a second variant answers a question the first cannot. Batch 0 comes first because it measures the one number every force in this report is waiting for.

Order	Part	Design / variant	Why it is on the list
0	Specimens + rig	Tensile 1BA (1.0, 2.0 mm), edge strips, Ø10 compression cylinder; test rig	SUNLU publishes no modulus. Every torque, force and return margin here scales with it; the specimens turn the 5.3–19.3 MPa bracket into one number.
1	Type A spring	A8 2-turn strip, 1.2 × 2.4 mm, sagitta 2.75 mm, with the A3 fit-check collars fused on (×2)	Strongest Type A spring inside the 20 % strain limit on the real index PIP; smooth to 100°; mesh-converged (0.45 %). With the collars as TPU: 2.81 N·mm at 90° at 19.3 MPa (1.75 at 12), 13 % below the rigid-end figure; fillet the foot–collar junction (21.5 % strain at 90°).
1	Type A spring	A8 same with collars, sagitta 3.00 mm (×1)	Fallback if the print stands proud: 0.60 mm clearance instead of 0.39 mm, −3 % torque.
1	Type A spring	A6 sliding leaf R28 / 1.4 / 6 mm (×2)	The only Type A layout that removes the A3's stretch problem by construction; 3.2 points of strain margin; the print locates the real joint axis, A6's largest unknown.
2	Type A spring	A6 R32 / 1.8 / 6 mm (×1)	Sweep optimum: about 3× the torque of R28 / 1.4, but at 19.8 % strain. Print once R28 works.
1	Type B joint	B5 crossed strip t2.0 / 45°, deep tendon (×2, deep phalanges)	Highest tendon capacity of any Type B joint in the FEA (30 N at 0°), and the only one above 2 N at the fingertip at 12 MPa.
1	Type B joint	B5 crossed strip t1.5 / 45°, as built (×2)	Round-1 winner: longest range (77.8°), least twist (4.8°), but buckles at 17 N of tendon tension. Brackets B5 against the t2.0 deep variant: range against compression capacity.
2	Type B joint	B4 rolling contact t1.0 / 60°, deep tendon (×1)	Best bearing result: at least 11 N at only 12 % strain when the contact solution stopped. Its lever falls as it rolls, which only a print can weigh against the capacity.
2	Type B joint	B8 knuckle-in-cup, y20 r1.25 deep (×1) and y00 r1.5 (×1)	Tests the pad bearing. The FEA confirms the seated knuckle carries load at low strain but cannot follow the sideways snap before it seats; only a print shows whether the cup self-centres.
3	Type B control	B3 dual leaf t1.5 / w3.0, as built (×1)	Optional: simplest Type B print and round-1 runner-up, as a reference for how much the crossed strip and the bearing improve on a plain leaf (B3 buckles at about 10 N).
—	Not printed	A3 (use the part you have), B1, B6, B9, lattice, the remaining deep copies	See section 6 for the reason each is set aside.

Full links to every file are in section 8. Every number in this report that depends on the TPU's stiffness is given at the working 12 MPa and across the 5.3–19.3 MPa bracket; see section 2.

The decision in three sentences

- **Type A is the safer route to a working finger.** Its pin carries the tendon's compression, the routing study puts the finger at 3.0–6.2 N at the fingertip, and both candidate springs (A8 strip, A6 sliding leaf) have converged or fully checked analyses.
- **Type B can work only with the B5 crossed strip and the deep tendon so far.** The round-2 FEA shows plain leaves (B3) buckling at about 10 N of tendon tension, a third of the hand estimate; strut and notch joints (B9, B1) run out of strain before 60°; the B5 t2.0 with the deep tendon carries 30 N.
- **The material test decides the rest.** At 5.3 MPa most Type B joints could not hold the finger straight and every force halves; at 19.3 MPa they gain 60 %. Batch 0 settles it.

2. The two joint concepts, and the material basis

The project carries two ways of making a finger joint. **Type A** keeps the existing hand: a PLA pin is the hinge, and a TPU part beside it only has to return the finger to straight. **Type B** removes the pin: the TPU flexure is the hinge, the return spring, and the only thing between the two phalanges when the tendon pulls. The difference decides what each design has to survive. A Type A spring is judged on range, strain and return torque. A Type B joint is also judged on twist, sideways stiffness, and above all on whether it can carry the tendon's compression without buckling, because a tendon that bends a joint also pushes its two halves together.

	Type A (pinned)	Type B (TPU joint, no pin)
What makes the hinge	A PLA pin; the axis is fixed	The TPU flexure itself; the axis is wherever it bends
What the TPU does	Only returns the finger (a spring beside the pin)	Is the joint, the spring, and carries the tendon's compression
Designs studied	A3 twin strap · A6 sliding leaf · A8 meander strip	Round 1: B1 hourglass, B3 dual leaf, B5 crossed strip, B6 multi-leaf, lattice · Round 2: B4 rolling, B8 knuckle, B9 strut, deep tendon
Tendon compression	Carried by the pin	Carried by the TPU — the problem round 2 addresses

Material basis

You asked for every result to use SUNLU TPU 90A, the filament in use. SUNLU publishes a density (1.23 g/cm³) and a tensile strength (40 ± 6 MPa), but **no stiffness**: neither its product page nor its store listing gives a modulus or a stress–strain curve, and the linked data-sheet PDF could not be retrieved from this environment (sunlu.com is outside its network allow-list). A modulus therefore cannot honestly be attributed to SUNLU. What this report does instead: every stiffness-dependent number (torques, forces, capacities, return margins) is quoted at a working 12 MPa and across a bracket of 5.3 MPa (a printed TPU 90A, Bambu Lab's data sheet) to 19.3 MPa (a bulk TPU 90A, Jabil's data sheet). Everything else (strains, ranges, collision angles, which variant ranks first) does not depend on the modulus at all, because every model here is a Neo-Hookean or linear solid loaded by an imposed motion or by a load whose effect scales with load/E. Where a pass/fail gate depends on the modulus, the verdict is shown at both ends of the bracket. The tensile specimens in the print list measure the real value; replacing one constant in scripts/sunlu_rescale.py then updates every number.

Quantity	Value used	Source
Density	1.23 g/cm³	SUNLU TPU 90A product page
Tensile strength	40 ± 6 MPa (not used by any model)	SUNLU TPU 90A product page
Young's modulus	**not published by SUNLU**; working 12 MPa, bracket 5.3–19.3 MPa	5.3: Bambu Lab TPU 90A sheet, printed, XY · 12: project working value · 19.3: Jabil TPU 90A (bulk)
Strain limit	20 % peak principal strain	Design assumption (not a measured fatigue limit)

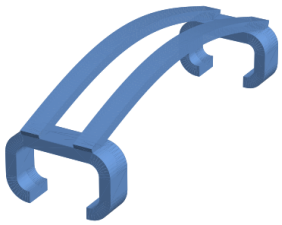
Every study was re-based from its own modulus (Type B and A3 12 MPa, A6 13 MPa, A8 19.3 MPa) by exact proportional scaling: scripts/sunlu_rescale.py → results/sunlu_rescale/headline_values.csv.

3. Type A: the spring beside the pin

All three Type A springs replace the A3 strap on the same index-finger PIP of the existing hand (the fit-check assembly): ends at the A3 collar positions, axis at the PLA pin. They differ in how they cope with the one geometric fact that dominates this joint: the straps sit about 4.4 mm above the pin, so as the finger bends the distance between the two attachment points first grows by about 1.3 mm (to about 34°) and only later shrinks.

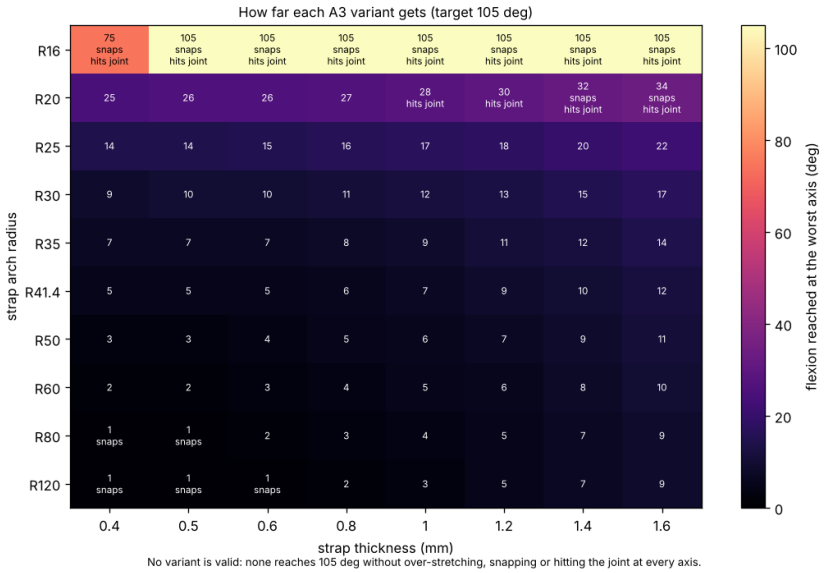
3.1 A3 twin strap (the current part)

A3 twin strap, 1.0 mm (current part)
11.4 × 35.2 × 11.0 mm



- **FEA is solid.** 6.103 N·mm/rad at neutral, converged to 0.94 % between the two finest meshes, reproduced to every digit from the GUI, and 9–20 % below the reduced-order model.
- **But the A3's layout cannot follow the finger.** A strap clamped at both ends has to stretch through the 0–60° window where the attachment points move apart. The planar elastica sweep of 480 curvature × thickness × width variants found **none** that reaches 105° without snapping, touching the joint or over-stretching, at any strain limit from 10 to 30 %. The as-built A3 has an equilibrium only to about 7°.
- **Caveat that a print settles in minutes:** the model's collars are rigid; the real A3 clips are C-shaped and may slide, which would act like the A6's sliding end. The existing printed A3 on the fit-check finger answers it: does it click or bottom out at 10–35°?
- **Decision:** the A3 is kept as the benchmark every other design is compared against, not developed further. No new A3 print is needed beyond the one you have.

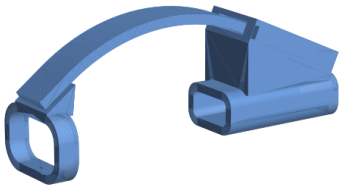
A3 result	Value	at 5.3 / 12 / 19.3 MPa
FEA rotational stiffness, pin-hinged (mesh 0.5 mm, 32 965 C3D10)	6.103 N·mm/rad	2.70 / 6.10 / 9.82
Mesh convergence, 1.0 → 0.5 mm	0.94 % change	—
Reduced-order model vs FEA	9–20 % stiffer (clamp-line dependent)	—
GUI run reproduces the scripted FEA	to every digit reported	—
Planar elastica sweep: 480 curvature × thickness × width variants	none valid at any strain limit 10–30 %	—
Furthest-reaching A3 (R20 / 0.8 mm) / as built (R41.7 / 1.0 mm)	27° / about 7° before snap or over-stretch	—



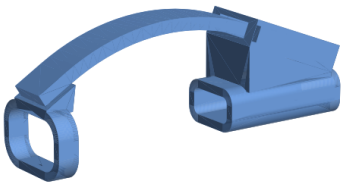
A3 parameter sweep: how far each curvature × thickness variant flexes before it snaps, touches the joint or over-stretches (results/a3_parameter_sweep).

3.2 A6 sliding-end leaf

A6 sliding leaf R28 / 1.4 / 6 mm
13.8 × 48.3 × 19.7 mm

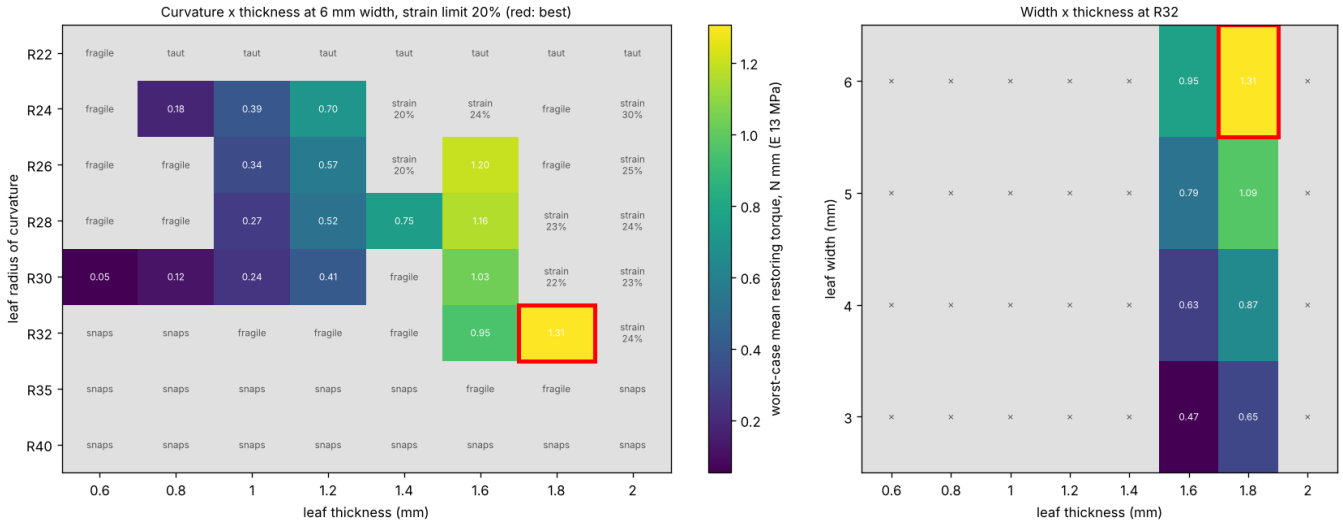


A6 sliding leaf R32 / 1.8 / 6 mm
13.8 × 48.3 × 18.7 mm



- **Why it exists:** one end of the leaf slides in a guide, which removes the change in end-to-end distance that defeats the A3. That is the fix for the A3's problem, by construction.
- **Sweep:** 8 curvatures × 8 thicknesses × 4 widths, each with the guide travel re-tuned, every cell judged at the worst of four assumed axis positions. The optimum, R32 / 1.8 mm / 6 mm, sits at 19.8 % strain against the 20 % limit; R28 / 1.4 mm / 6 mm has 3.2 points of strain margin and the widest travel window.
- **All 24 as-built checks pass** (six designs × four axis positions).
- **Weakness:** even the best A6 returns with about a fifth of the A3's neutral stiffness, and the valid region is patchy (0.5–1 mm travel windows), so print and axis tolerance matter. A6 FEA was **deliberately skipped** in this round: its biggest unknown is the real axis position, which a quick print settles and a simulation cannot.

A6 design	Peak strain (worst axis)	Closest to axis	Head reaches stop	Torque at 105°, 5.3 / 12 / 19.3 MPa
R32 / 1.8 / 6 mm (sweep optimum)	19.8 %	4.18 mm	53–62°	0.93 / 2.10 / 3.38 N·mm
R28 / 1.4 / 6 mm (with margin)	16.8 %	4.26 mm	50–58°	0.29 / 0.66 / 1.05 N·mm



A6 curvature × thickness sweep at 6 mm width (results/a6). Stars mark the optimum; hatched cells fail.

3.3 A8 meander strip



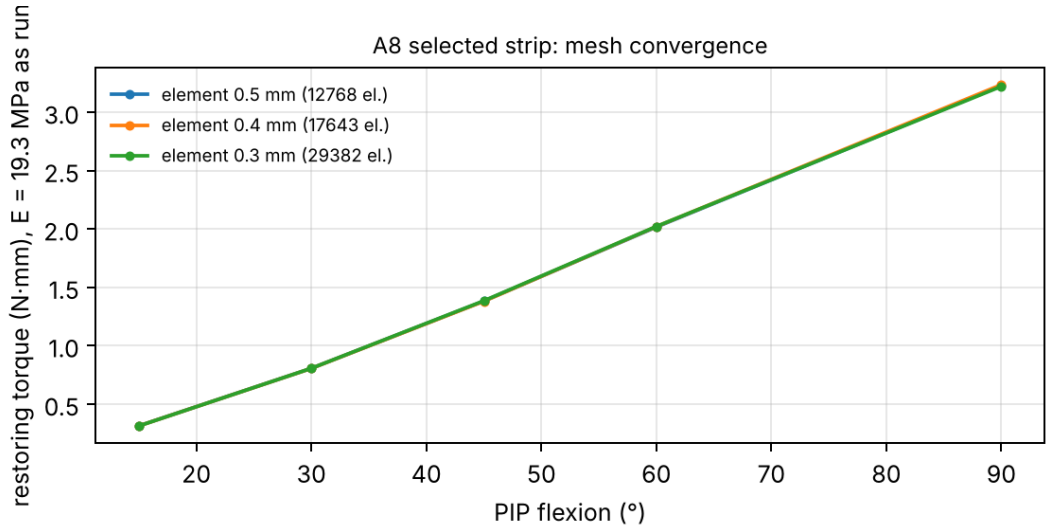
- **Why it exists:** a serpentine can lengthen and shorten in plan, so it accommodates the attachment points moving apart, which the A3 arch cannot.
- **Part 1** (your three SolidWorks exports, finger inferred from the A8 rings): 2/3/4 turns × spine curvature by contact FEA. It found the drawn section too weak (under 0.5 N·mm) and unprintable with a 0.4 mm nozzle: the 3-turn plan's lobe tips are 0.62 mm in radius.
- **Part 2** (the real index PIP from the fit-check assembly, printable 1.2 mm true width): 2 turns beats 3 at every height; strain, not stiffness, is the limit, because wrapping the R 5.2 cams costs about h/2R by itself; the best curvature is the flattest arch that still clears the real middle phalanx by 0.25 mm. Selected: 2 turns, 1.2 × 2.4 mm, sagitta 2.75 mm.
- **The one weak spot:** 26 % strain where the strip enters the collar at 90° with rigid ends, against 16.5 % in the strip. With real (TPU) collars it falls to 21.5 % (next subsection), still just over the limit at 90° and 12.8 % at 60°. It is where a print will crack first: blend the junction with a 1–2 mm fillet.

A8 selected strip: 2 turns, 1.2 × 2.4 mm, sagitta 2.75 mm	**5.3 MPa**	**12 MPa**	**19.3 MPa (as run)**
Restoring torque at 30°	0.22 N·mm	0.50 N·mm	0.81 N·mm
Restoring torque at 60°	0.55 N·mm	1.26 N·mm	2.02 N·mm
Restoring torque at 90°	0.89 N·mm	2.01 N·mm	3.24 N·mm
Peak strain at 90° (interior / at collar)	16.5 % / 26 %	same	same
Clearance to real Index_PP / Index_MP	1.06 / 0.39 mm	same	same

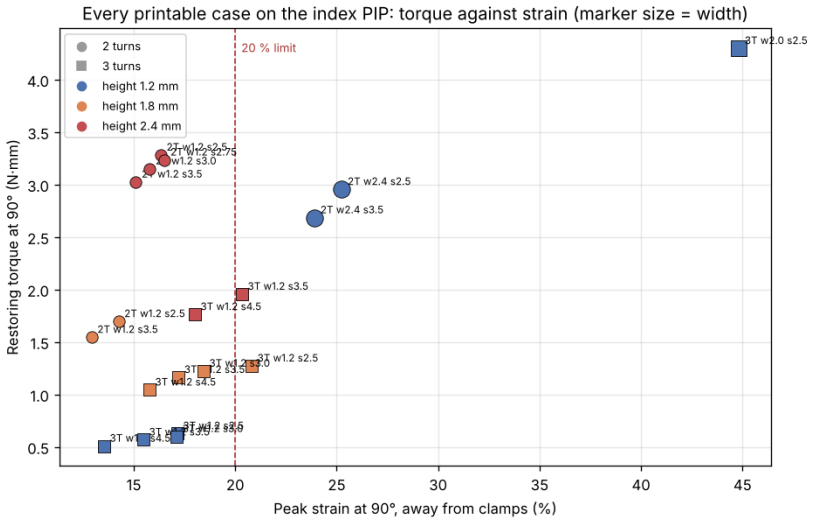
Mesh convergence of the selected strip (new in this round)

0.5 mm (12768 elements): torque at 90° 3.228 N·mm, peak interior strain 15.6 %; 0.4 mm (17643 elements): torque at 90° 3.236 N·mm, peak interior strain 16.5 %; 0.3 mm (29382 elements): torque at 90° 3.221 N·mm, peak interior strain 16.1 %. Between the study's 0.4 mm mesh and the finest, torque at 90°

changes by +0.5 % and peak strain by +2.8 %. That is inside the 2 % criterion the project uses (Phase 3 criterion 3), so the A8 recommendation stands on a converged mesh.



Torque–angle of the selected A8 strip at three mesh sizes (results/a8_mesh_convergence; 0.4 mm is the original study's mesh).



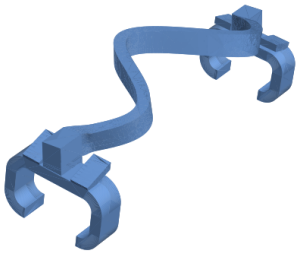
A8 printable cases on the real index PIP: torque at 90° against peak strain; the dashed line is the 20 % limit (results/a8_index_pip_printable).

3.4 Type A tendon routing

- From the hand assembly: the finger holds **3.0–3.4 N** at the fingertip with guides flush to the palmar face, **4.3–6.2 N** with guides standing proud, limited by servo stroke (placeholder MG995 at half stall, 47 N, 31.4 mm).
- How far below the joint axis the tendon runs decides almost everything; where along the phalanx the guide sits changes force by at most 20 %.
- **The DIP only flexes to 40–51°** in the CAD (human: 80–90°), confirmed by an exact solid intersection. If that is not intended it is the first thing to change in the Type A hand.
- The pin carries the tendon's compression, so none of the Type B buckling problem applies.

The A8 strip with collars (new in this round)

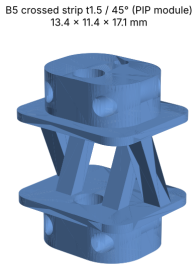
A8 strip with A3 fit-check collars, sagitta 2.75 mm
11.7 × 35.1 × 12.9 mm

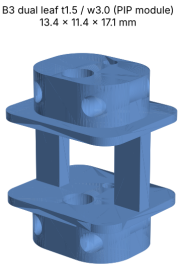


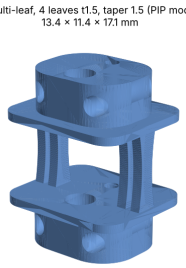
- Both A8 analyses held the strip's ends rigidly and left the collars out: Part 1 kept only the strip from your exports (whose rings were sized for a finger inferred from the exports), and the printable strip was exported as the strip alone. For this round the printable strip was fused to the collars of the A3 0.8 mm fit-check strap, which were designed for the real Index_PP and Index_MP, at exactly the positions where the A3 straps leave them (scripts/a8_collared_geometry.py). Because the strip's ends stop where the collars stop and float 0.05–0.4 mm above them, each end was given a foot running back along the collar's top and into its wall. The result is one 313 mm³ TPU part, then re-run in the A8 contact FEA with the collars as TPU instead of rigid ends.
- **Sagitta 2.75 mm, with collars** (reached 100°): torque at 90° 2.81 N·mm at 19.3 MPa; peak strain at 90° 14.7 % in the strip, 21.5 % at the strip–collar junction, 10.6 % in the collars. Without collars (rigid ends) the same strip gave 3.24 N·mm, 16.5 % in the strip and 26.4 % at its rigidly held ends.
- How the collars are held in this FEA: on every collar face that faces its phalanx. The A3 collars' inner faces sit 0.36–1.0 mm off the bone in the CAD rather than on it, so this represents a collar gripped on the phalanx (snug, or bonded), flexible everywhere else. If the printed collars are a loose fit, the joint between collar and bone will be softer still.

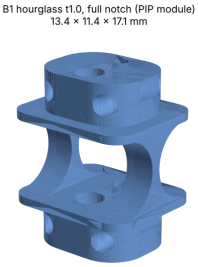
4. Type B round 1: the TPU joint as a spring

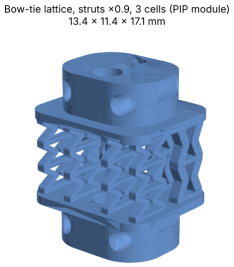
Five families, 55 variants, each simulated as a whole flex zone bolted between the phalanges with a full nonlinear flexion sweep and three off-axis load cases (twist, sideways bend, axial). 42 leaf variants and 4 of 13 lattice variants were run at the PIP; the winners were checked at the MCP and DIP. A mesh check (2 vs 3 elements through) moved every selection metric by under 1 %. The 9 remaining lattice runs were **deliberately skipped** this round: the four run all reach 20 % strain by 10–20°, the trend with strut thickness is monotonic, and the missing variants change cell count, not slenderness.





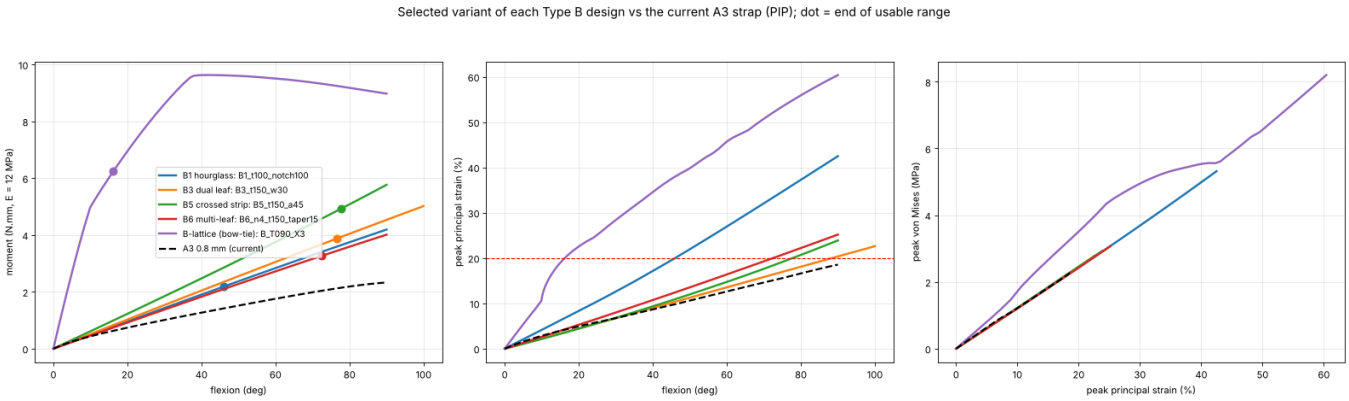




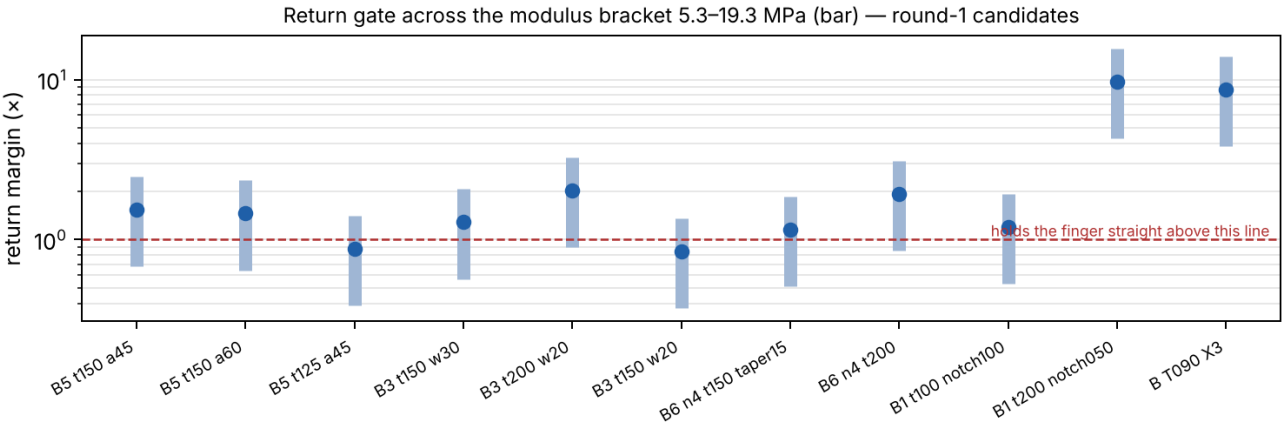


Family	Variant	Range (°)	Limited by	Strain @60°	Tendon force (N)	Return margin, 5.3/12/19.3 MPa	Twist (°)
B5 crossed	B5_t150_a45	77.8	strain 20 %	14.7 %	2.61	0.68 / 1.54 / 2.48	4.8
	B5_t150_a60	74.6	strain 20 %	15.3 %	2.44	0.64 / 1.46 / 2.34	3.3
	B5_t150_a75	75.4	strain 20 %	15.1 %	2.43	0.60 / 1.36 / 2.19	2.5
	B5_t200_a45	59.1	strain 20 %	20.3 %	4.72	1.65 / 3.74 / 6.01	3.2
	B5_t125_a45	77.9	collision	12.8 %	1.50	0.39 / 0.88 / 1.41	6.1
B3 dual leaf	B3_t150_w30	76.6	collision	13.5 %	2.05	0.57 / 1.29 / 2.07	16.1
	B3_t200_w20	72.5	strain 20 %	16.3 %	3.04	0.90 / 2.03 / 3.26	9.8
	B3_t200_w30	67.5	strain 20 %	17.7 %	4.31	1.36 / 3.09 / 4.96	7.3
	B3_t150_w20	76.5	collision	12.4 %	1.34	0.37 / 0.84 / 1.36	21.8
B6 multi-leaf	B6_n4_t150_taper15	72.5	strain 20 %	16.4 %	1.72	0.51 / 1.15 / 1.85	15.0
	B6_n4_t200	64.2	strain 20 %	18.6 %	2.59	0.86 / 1.94 / 3.12	12.7
B1 hourglass	B1_t100_notch100	46.0	strain 20 %	27.0 %	1.14	0.53 / 1.20 / 1.93	5.4
	B1_t200_notch050	17.9	strain 20 %	68.7 %	3.64	4.30 / 9.73 / 15.65	1.1
Lattice	B_T090_X3	16.2	strain 20 %	45.7 %	1.38	3.85 / 8.72 / 14.03	30.6
	B00_baseline	20.2	strain 20 %	–	2.09	5.33 / 12.07 / 19.41	21.7

Return margin = restoring moment at 20° divided by the moment of the distal finger's weight; at least 1 to hold the finger straight. It scales with E, so the bracket decides eligibility for the softer variants.



Round-1 FEA: the selected variant of each family against the A3 benchmark (TYPE B/analysis/REPORT.md).



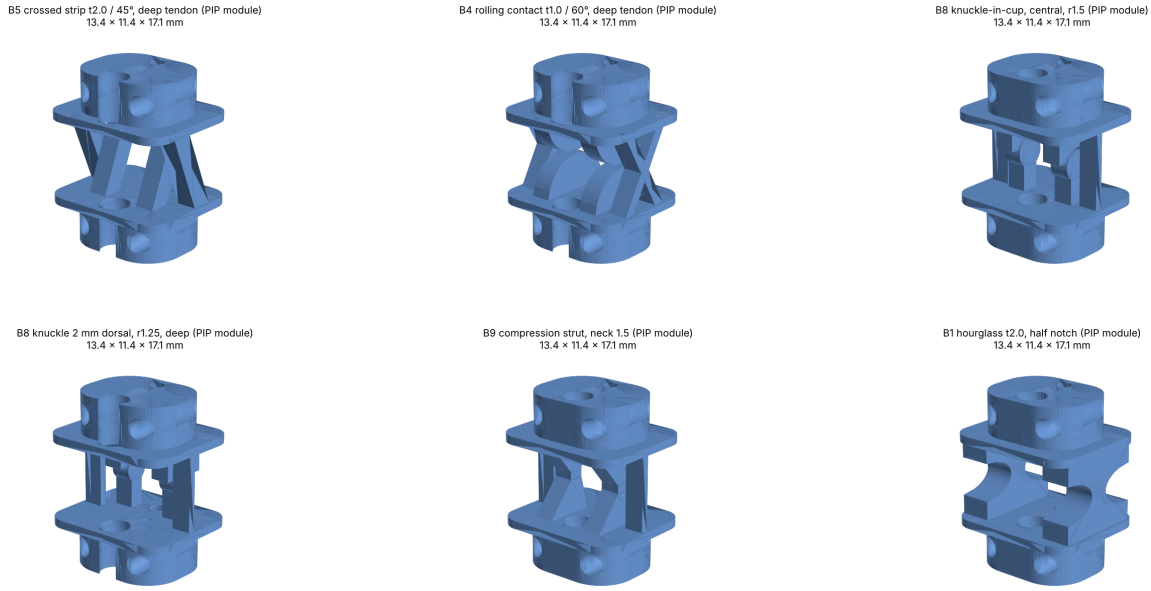
Return gate across the 5.3–19.3 MPa bracket. Bars that cross the red line change verdict with the modulus.

What round 1 did not test

Round 1 bends each joint with a pure moment and lets it translate freely, which is the right test of a spring and the wrong test of a joint that has to carry a tendon. At the MCP none of the winners passes the return gate at 12 MPa (0.38–0.63×), so the MCP needs the thicker grading of REPORT.md §5.2 whatever else is chosen. At the DIP all pass the return gate (3.5–4.0×), and the B3, B5 and B6 winners reach the full 80°; the B1 winner, checked at the DIP in the last round-1 commit, is strain-limited at 52.7°, the same throat problem it has at the PIP (46°) and MCP (50.5°). The tendon-routing study (docs/11) then estimated, by Euler buckling, that every round-1 leaf buckles at a fraction of the servo's tension, which is why round 2 exists and why section 5 measures that compression directly.

5. Type B round 2: carrying the tendon's compression

Round 2 adds three ways of carrying the tendon's compression so the TPU only has to spring the joint back: **B4** rolling lobes, **B8** a knuckle in a cup (central or 2 mm dorsal), **B9** a stocky compression strut with one neck; and a **deep tendon** option that moves the tendon channel as far palmar as the section allows (PIP 1.93 → 3.40 mm), so the same fingertip force needs less tension. Until now these had only hand estimates. This round simulated the TEST_PLAN shortlist (rows 1–11) by nonlinear FEA.



Method

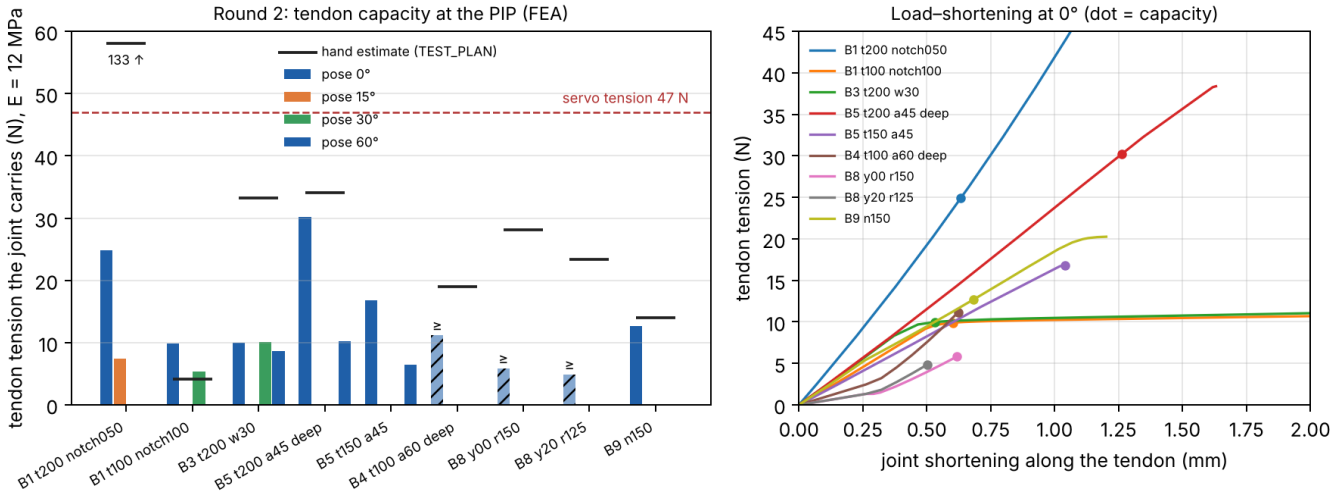
- Same flex zone and attachment as round 1 (proximal collar fixed, distal collar a rigid body).
- **Bend** the joint to the pose (0° and 60°) with its translations free, then **hold the angle** and ramp a force along the tendon's line, from its distal entry towards its proximal exit, plus a 1 % sideways push so a symmetric joint can find its buckling mode.
- Bearing faces (B4 lobes, B8 knuckle and cup, 0.3 mm apart as printed) carry a surface-to-surface contact pair, penalty 10 E/h.
- Capacity is the first of: the solver finds no equilibrium (collapse), the axial stiffness falls below 10 % of its initial value (buckling), or peak strain reaches 20 %.
- Graded meshes (two quadratic elements through each leaf, 0.5 mm in bearings, 0.8 mm in the collars): a uniform leaf-size mesh needed over 12 GB in SPOOLES and was killed.
- Verified first: a committed round-1 run (B3_t200_w30) re-run here reproduced its moments to five figures and its collision angle to 0.02°.

Results

#	TEST_PLAN row	FEA run	Capacity @0° (N)	Capacity @60° or as noted (N)	Limited by (0°)	Hand est. (N)	Fingertip force @0°, 5.3/12/19.3 MPa (N)
1	B1_t200_notch050	B1_t200_notch050	24.9	0 (over 20 % from the bend); 7.4 at 15°	20 % strain	133.0	0.47 / 1.07 / 1.72
2	B1_t200_notch050_deep	B1_t200_notch050	24.9	0 (over 20 % from the bend); 7.4 at 15°	20 % strain	133.0	0.83 / 1.89 / 3.04
3	B3_t200_w30	B3_t200_w30	10.0	8.6	stiffness loss (buckling)	33.2	0.19 / 0.43 / 0.69
4	B3_t200_w30_deep	B3_t200_w30	10.0	8.6	stiffness loss (buckling)	33.2	0.33 / 0.76 / 1.22
5	B5_t200_a45_deep	B5_t200_a45_deep	30.2	10.3	20 % strain	34.1	1.01 / 2.29 / 3.69

#	TEST_PLAN row	FEA run	Capacity @0° (N)	Capacity @60° or as noted (N)	Limited by (0°)	Hand est. (N)	Fingertip force @0°, 5.3/12/19.3 MPa (N)
6	B4_t100_a60_deep	B4_t100_a60_deep	≥ 11.1	–	lower bound: contact diverged (guided)	19.0	≥ 0.37 / 0.85 / 1.36
7	B8_y00_r150	B8_y00_r150	≥ 5.8	–	lower bound: contact diverged (guided)	28.1	≥ 0.11 / 0.25 / 0.40
8	B8_y20_r125	B8_y20_r125	≥ 4.9	–	lower bound: contact diverged (guided)	23.4	≥ 0.19 / 0.43 / 0.69
9	B8_y20_r125_deep	B8_y20_r125	≥ 4.9	–	lower bound: contact diverged (guided)	23.4	≥ 0.26 / 0.59 / 0.94
10	B9_n150	B9_n150	12.7	0 (over 20 % from the bend)	20 % strain	14.0	0.24 / 0.55 / 0.88
11	B9_n150_deep	B9_n150	12.7	0 (over 20 % from the bend)	20 % strain	14.0	0.43 / 0.96 / 1.55
+	B5_t150_a45 (round-1 winner)	B5_t150_a45	16.8	6.4	limit point (collapse)	–	0.32 / 0.73 / 1.17
+	B5_t150_a45_deep	B5_t150_a45_deep	16.8	6.4	limit point (collapse); same deck as the as-built run	–	0.56 / 1.28 / 2.05
+	B1_t100_notch100 (B1 round-1 winner)	B1_t100_notch100	9.9	5.3 at 30°	20 % strain	–	0.19 / 0.43 / 0.69
+	B1_t100_notch100_deep	B1_t100_notch100_deep	9.9	5.3 at 30°	20 % strain; same deck as the as-built run	4.2	0.33 / 0.75 / 1.21

Capacity = the tendon tension at which the joint first buckles, collapses or reaches 20 % strain while held at the pose, at E = 12 MPa (scales with E). Fingertip force = capacity × tendon lever ÷ 44.8 mm, the PIP-to-hook distance of the TEST_PLAN rig; the spring's own moment is neglected. Deep copies whose flexure is unchanged share their as-built run; only the lever differs.



Round-2 FEA: tendon capacity per shortlisted joint, against the hand estimates in TEST_PLAN.md and the servo's tension.

What the round-2 FEA shows

- **The hand estimates were optimistic, by up to 5×.** B3_t200_w30 buckles sideways at about 10 N (estimate 33 N); B1_t200_notch050 reaches 20 % strain at 25 N (estimate 133 N, an Euler load the notch never gets near because it runs out of strain first). The estimates assumed fixed-fixed buckling; the real joint can also sway sideways, which needs a quarter of the load.
- **B5 crossed strip with the deep tendon is the strongest Type B joint:** 30 N at 0° (estimate 34 N) and 10 N at 60°, both limited by strain rather than buckling. Crossing the strips braces the joint against the sideways sway that defeats parallel leaves. The thinner round-1 winner, B5_t150_a45, buckles at 17 N at 0° and reaches 20 % strain at 6.4 N at 60°: thickness buys compression capacity and costs range.

- **B1 hourglass, both variants, is strain-limited throughout.** The B1 round-1 winner B1_t100_notch100 reaches 20 % strain at 9.9 N of tendon tension at 0° and at 5.3 N at 30°, as built and with the deep tendon alike (at 30° the bend's own strain dominates). B1_t200_notch050 carries 24.9 N at 0° but only 7.4 N at 15°, inside its 18° range.
- **Bending uses up the strain budget.** At 60° B1_t200_notch050 is already at 68 % strain and B9_n150 at 37 % from the bend alone, before any tendon load, so neither can be a finger joint over that range. B5 and B3 lose about two thirds of their capacity between 0° and 60° for the same reason.
- **The B8 bearing carries load at low strain once seated, but could not be resolved to its limit.** Its 0.75 mm return leaves buckle sideways at about 0.3 N, so under a free load the joint snaps across its 0.3 mm gap before the knuckle seats. Driven along the tendon line instead (guided), the seated knuckle carried at least 5.8 N (y00, 16 % strain) and 4.9 N (y20, 6 % strain) before CalculiX's contact iterations diverged. These are lower bounds. At 60° neither B8 nor B4 could be resolved: the contact solution diverged during the bend (B8_y20) or on the first load increment (B8_y00, B4). Whether a real cup self-centres through the snap is the question its print answers.
- **B4 rolling contact (deep) is the most promising bearing:** guided, it carried at least 11.1 N at 12 % strain at 0° (estimate 19 N) when the contact solution stopped, well short of its strain limit. Its catch is kinematic, not structural: the rolling contact moves palmar, so the tendon lever falls from 3.40 mm to about 1.4 mm by 60°.
- **Even the best Type B joint holds less at the fingertip than Type A.** B5 t2.0 deep converts 30 N of tension into about 2.3 N at the 44.8 mm hook at 12 MPa (1.0–3.7 N across the modulus bracket), against 3.0–6.2 N for the pinned Type A finger, whose limit is the servo's stroke, not the joint.

6. Why these designs are being carried forward

Why Type A springs are printed first

- The pin takes the tendon's compression, so a Type A spring only has to return the finger. That is a much easier job, and it is fully analysed: A8 by converged contact FEA on the real joint geometry, A6 by a sweep plus 24 as-built checks.
- **A8 2-turn strip, sagitta 2.75 mm.** Chosen in the A8 study by explicit rules: clear the real phalanx solids by 0.25 mm, no torque drop in 0–100°, at most 20 % strain at 90°, then most torque. Two turns beat three at every height because the 2-turn strip does not snap on the real joint. Height, not width, buys torque: wider strips bend in plane and exceed the strain limit. This round confirmed the mesh: torque at 90° moves 0.45 % from 0.4 to 0.3 mm elements. Print two, because one will be used up in a destructive test.
- **A8 sagitta 3.00 mm as well.** The 2.75 mm arch clears the real middle phalanx by 0.39 mm; a print that stands proud by 0.2 mm would touch. The 3.0 mm arch keeps 0.60 mm for 3 % less torque. Printing both in the same session settles it at once.
- **A6 R28 / 1.4 mm before R32 / 1.8 mm.** The A6 exists because the A3 cannot work as laid out: the elastica sweep found none of 480 A3 variants valid, because a strap clamped at both ends must stretch while the attachment points move apart. A sliding end removes that by construction. R32 / 1.8 is the sweep optimum but sits at 19.8 % strain; R28 / 1.4 keeps 3.2 points of margin and the widest travel window, so it is the one most likely to work first time. Its print also locates the real joint axis, which every A6 result currently guesses at four positions. That is why A6 FEA was skipped.
- **A3 is not reprinted.** It is the benchmark (6.10 N·mm/rad, converged), and the one question the model leaves open (do the C-clip collars slide and rescue it?) is answered by flexing the A3 you already have and noting whether it clicks at 10–35°.

Why B5 is the Type B design carried forward, in two variants

- Round 1 ranked B5_t150_a45 first on range, twist and robustness to weighting. Round 2 ranked B5_t200_a45 with the deep tendon first on tendon capacity. They are the same design at two thicknesses, and thickness trades range against compression capacity, so printing both measures that trade directly instead of trusting either model alone.
- The deep tendon matters as much as the flexure: at the same tension it gives 1.76× the fingertip force (3.40 mm against 1.93 mm lever), and it costs nothing in the flexure. It does need the deep-offset phalanges.

Why B4 and B8 are printed although their FEA is incomplete

- B4 deep is printed for the same reason as B8 below, and with better numbers: it carried at least 11.1 N at 12 % strain at 0° before the contact solution stopped. The question its print answers is whether the falling lever (3.40 → about 1.4 mm by 60°) leaves enough fingertip force through the range.
- B8 is the other round-2 design whose failure mode the FEA could not reach: the seated knuckle was still at 6–16 % strain when the contact solution diverged, so its capacity is a lower bound, not a limit. The open question is mechanical rather than numerical: does the cup catch and centre the knuckle when the thin leaves sway? One print of each pad position answers that in minutes on the rig, and the y20 deep copy has the longest tendon lever of anything designed (5.4 mm at 0°).

What is set aside, and why

- **B1 hourglass:** the notch concentrates bending. The round-1 winner, t1.0 / full notch, is strain-limited at 46° (PIP), 50.5° (MCP) and 52.7° (DIP), and under tendon load carries 9.9 N at 0° and 5.3 N at 30°, against B5 deep's 30 N and 10 N. The t2.0 / half notch reaches 20 % strain by 18° from the bend alone; its 25 N at 0° drops to 7.4 N at 15°. High capacity at 0° does not help a joint that cannot bend.
- **B9 compression strut:** 12.7 N at 0°, close to its estimate, but 37 % strain at 60° from the bend alone; the neck cannot take the joint through a finger's range.
- **B3 dual leaf:** buckles sideways at about 10 N even at 2 mm, and is the softest in twist (16° under a 1 N pinch). Kept only as an optional plain-leaf control.
- **B6 multi-leaf:** behaves like a split B3 with no torsional gain, and was not shortlisted for round 2.
- **Lattice:** 20 % strain by 10–20° in every variant run; the missing variants change cell count, not strut slenderness. Skipped by agreement.

- **B4 as built and the other deep copies:** as built, B4's tendon lever falls to 0.9 mm by 60° because the contact moves palmar, so only the deep copy is printed. Deep copies of B1, B3, B6, B9 and the lattice inherit their flexure's limits above.

7. What is still unknown, and what each print will settle

Unknown	Why it matters	What settles it
TPU modulus (SUNLU publishes none)	Every force, torque, capacity and return margin scales with it; at 5.3 MPa several Type B joints fail the return gate	Batch 0 tensile specimens; then edit E_SUNLU_MPA in scripts/sunlu_rescale.py and rebuild this report
Strain limit (20 % is an assumption)	Decides A6's optimum, A8's height, and every Type B range	Cycling a printed spring (TEST_PLAN §4 step 5)
Real PIP axis position for A6	Every A6 result is a worst case over four guesses	The A6 R28 / 1.4 print on the fit-check finger
Whether the B8 cup self-centres	FEA capacity is a lower bound; the joint may wobble sideways before seating	B8 prints on the rig, TEST_PLAN §4 step 3
Print accuracy of thin features	Stiffness goes as thickness cubed: a 1.0 mm leaf printed at 1.1 mm is 33 % stiffer	Calipers on every part before testing; reject beyond ±10 %
Whether the A3 C-clips slide	Could make the as-built A3 better than the elastica model says	Flex the existing A3 print and note any click at 10–35°
DIP flexion stop at 40–51° in the Type A CAD	Human DIP reaches 80–90°	A CAD decision, not a test: change the DP/MP geometry if unintended

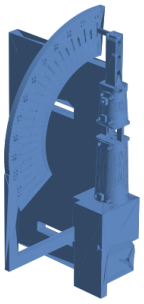
8. How to print each part

General rules first, then the part-by-part specification. These follow TEST_PLAN.md §2 and common practice for a soft (90A) TPU; they are not results of this project's analyses, except where a part's geometry (a gap, a thin wall) is quoted.

- **Dry the TPU** before printing (about 50 °C for 4–6 h, or per the spool) and keep it in a dry box while printing. Wet TPU strings, foams and prints softer.
- **Slow and steady:** about 15–30 mm/s, minimal or no retraction; direct drive is easier with 90A. Use the nozzle and bed temperatures on the spool (SUNLU's sheet could not be read here). A glue-stick release layer helps on PEI or glass, which TPU can bond to strongly.
- **100 % infill, 0.4 mm line width, 0.2 mm layers**, and the same settings, spool and session for every part of a batch, including the tensile specimens.
- **Orientation:** a print is weakest between layers, and a flexure's strain runs along its leaf. So lay leaves, strips and necks **lengthwise in the XY plane**, so their thickness is set by perimeters (accurate) rather than by layer count (in 0.2 mm steps). Stiffness goes as thickness cubed: +0.1 mm on a 1.0 mm leaf is +33 %.
- **Check every part:** calipers on every leaf, strip or neck at three points; feeler gauge through every printed-in-place gap; weigh each part. Reject anything beyond ±10 % of nominal.

Part	Material, qty	Orientation on the bed	Supports	Watch for / check after
Tensile 1BA, 1.0 / 2.0 mm	TPU, 3 each	Flat on the bed	None	Same session as the joints. Pull at constant speed; record force–extension to at least the strain the parts see (20 %).
Tensile strip on edge, 1.0 / 2.0 mm	TPU, 3 each	Standing on its long edge, 6 mm tall (laid down as a leaf is)	None; brim if it wobbles	Thin tall wall: slow down. Compare its modulus with the flat 1BA to see the effect of print direction.
Compression cylinder Ø10 × 10	TPU, 3	On a flat end	None	—
A8 strip with collars, s2.75 (and s3.00)	TPU, 2 (+1)	Collars' open (palmar) side up or on their side so the strip's 1.2 mm width lies in XY; the strip arches over the joint	Under the arch and the collar overhangs; easy-peel interface	Strip width is exactly 3 lines: check 1.2 mm ±0.1. Check it clips onto Index_PP / Index_MP; the collar–foot junction is the likely crack point.
A6 R28 / 1.4 and R32 / 1.8	TPU, 2 + 1	On its side, leaf curve in the XY plane	Under the guide only	Two bodies printed in place with 0.25 mm sliding clearance: free the slider after printing; a fused slider is a reject. Leaf thickness 1.4 / 1.8 mm ±10 %.
Type B modules (B5, B3)	TPU, per list	On its side, lateral axis vertical, so every leaf or strip lies in XY	Under the bridging upper lanes only; support blockers elsewhere	Leaf/strip thickness at 3 points (1.5 or 2.0 mm). Never print standing on a tab end.
B4 rolling contact (deep)	TPU, 1	On its side, as above; lobe profiles then lie in XY	Under the lanes only	0.3 mm lobe gap must be open: 0.25 mm feeler gauge. Elephant's foot can close it near the bed.
B8 knuckle-in-cup	TPU, 1 each	On its side; knuckle and cup circles lie in XY	Under the lanes only	0.3 mm gap as for B4. Return leaves are 0.75 mm (PIP), 0.61 mm (DIP): under two line widths, so the slicer may thin or single-wall them; measure.
Type B phalanges (as built / deep)	PLA, 1 set per module	Standard	As needed	3+ perimeters. Deep-tendon modules need the deep phalanges: their tendon groove is closed by the PLA socket wall.
Rig base, load adapter, pulley	PLA, 1 each	Rig base on its side plate	As needed	4 perimeters on the base (the protractor stands vertical); brim if it wobbles. Pulley on a 623ZZ bearing and M3 bolt.

Single-joint test rig (assembly)
89.0 × 74.0 × 148.8 mm



9. Files to print — links

Every link points at the file on branch claude/modest-cray-kbb75b of Melle2006/BARNES_UI (GitHub: open, then Download raw file). STL is for the slicer; STEP is the same part for CAD or slicers that accept it. Print every TPU part of one batch from the same dried spool in one session, at 100 % infill, 0.4 mm line width, 0.2 mm layers, with leaves lying lengthwise in the XY plane (modules on their side), as TEST_PLAN.md §2 sets out; measure every leaf, strip or neck at three points before testing and reject parts more than ±10 % off nominal.

Batch 0 — measure the material first (TPU, same spool as everything else)

Part	Print	Files
Tensile dog-bone 1BA, 1.0 mm and 2.0 mm, printed flat	TPU ×3 each	tensile_1BA_t1.0_flat.stl tensile_1BA_t2.0_flat.stl
Tensile strip, 1.0 and 2.0 mm, standing on its edge (laid down as a leaf is)	TPU ×3 each	tensile_strip_t1.0_edge.stl tensile_strip_t2.0_edge.stl
Compression cylinder Ø10 × 10 mm	TPU ×3, on its flat end	compression_d10_h10.stl

Test rig (PLA)

Part	Print	Files
Rig base (clamps to a table edge; protractor)	PLA, on its side plate, 4 perimeters	rig_base.stl rig_assembly.step
Load adapter (hook arm at 30 / 37.5 / 44.8 mm)	PLA, 3+ perimeters	load_adapter.stl
Tendon pulley (623ZZ bearing on an M3 bolt)	PLA	pulley.stl
Type B phalanges (as-built tendon)	PLA, one set per as-built module tested	02_Phalanx1.stl 04_Phalanx2.stl 06_Phalanx3.stl
Type B phalanges (deep tendon)	PLA, one set per deep module tested	02_Phalanx1.stl 04_Phalanx2.stl 06_Phalanx3.stl

Type A springs for the existing hand (index PIP)

Part	Print	Files
A8 2-turn strip with A3 fit-check collars, sagitta 2.75 mm (selected)	TPU ×2, one piece	A8_PIP_2turn_printable_w1.2_h2.4_s2.75_collared_study_frame.stl A8_PIP_2turn_printable_w1.2_h2.4_s2.75_collared_study_frame.step A8_PIP_2turn_printable_w1.2_h2.4_s2.75_collared_Assem3_frame.step
A8 with collars, sagitta 3.00 mm (fallback if prints come out proud)	TPU ×1, one piece	A8_PIP_2turn_printable_w1.2_h2.4_s3.00_collared_study_frame.stl A8_PIP_2turn_printable_w1.2_h2.4_s3.00_collared_study_frame.step
A8 strip alone, s2.75 / s3.00 (if you attach it your own way)	TPU	A8_PIP_2turn_printable_w1.2_h2.4_s2.75_IndexPP_frame.stl A8_PIP_2turn_printable_w1.2_h2.4_s3.00_IndexPP_frame.stl
A6 sliding leaf R28 / 1.4 / 6 mm (print first: strain margin)	TPU, both bodies printed in place (0.25 mm sliding clearance), ×2	A6_PIP_SlidingLeaf_R28_t1.4_w6.STL A6_PIP_SlidingLeaf_R28_t1.4_w6.STEP
A6 sliding leaf R32 / 1.8 / 6 mm (sweep optimum, at the strain limit)	TPU, both bodies in place, ×1	A6_PIP_SlidingLeaf_R32_t1.8_w6.STL A6_PIP_SlidingLeaf_R32_t1.8_w6.STEP
A3 twin strap (benchmark: test the print you already have)	no new print needed	A3_PIP_twin_strap_1.0mm.step

Type B joint modules (TPU). Rig tests use the PIP module; MCP and DIP modules are for a whole finger

Part	Print	Files
B5 crossed strip t2.0 / 45°, deep tendon	TPU ×2 (PIP); with the deep phalanges	03_PIP_module.stl 03_PIP_module.step 01_MCP_module_with_flange.stl 05_DIP_module.stl finger_assembly.step
B5 crossed strip t1.5 / 45°, as built	TPU ×2 (PIP); with the as-built phalanges	03_PIP_module.stl 03_PIP_module.step 01_MCP_module_with_flange.stl 05_DIP_module.stl finger_assembly.step
B4 rolling contact t1.0 / 60°, deep tendon	TPU ×1 (PIP); check the 0.3 mm lobe gap is open	03_PIP_module.stl 03_PIP_module.step 01_MCP_module_with_flange.stl 05_DIP_module.stl finger_assembly.step
B8 knuckle 2 mm dorsal, r1.25, deep tendon	TPU ×1 (PIP); check the 0.3 mm gap is open	03_PIP_module.stl 03_PIP_module.step 01_MCP_module_with_flange.stl 05_DIP_module.stl finger_assembly.step
B8 knuckle central, r1.5, as built	TPU ×1 (PIP); check the 0.3 mm gap is open	03_PIP_module.stl 03_PIP_module.step 01_MCP_module_with_flange.stl 05_DIP_module.stl finger_assembly.step
B3 dual leaf t1.5 / w3.0, as built (optional control)	TPU ×1 (PIP)	03_PIP_module.stl 03_PIP_module.step 01_MCP_module_with_flange.stl 05_DIP_module.stl finger_assembly.step

Appendix: where every result lives

Study	Results	Script
A3 FEA, mesh study	results/a3_mesh_study/	scripts/a3_mesh_study.py
A3 reduced-order vs FEA	results/a3_rom_vs_fea.json	scripts/a3_rom_vs_fea.py
A3 parameter sweep	results/a3_parameter_sweep/	scripts/a3_parameter_sweep.py
A3 GUI run	results/a3_gui_run/	docs/manual/running-a-simulation.md
A6 sweep and checks	results/a6/	scripts/a6_parameter_sweep.py
A8 meander study (part 1)	results/a8_meander_study/	scripts/a8_meander_study.py
A8 printable study (part 2)	results/a8_index_pip_printable/	scripts/a8_printable_report.py
A8 mesh convergence (new)	results/a8_mesh_convergence/	scripts/a8_meander_study.py
Type A tendon routing	results/tendon_routing/	scripts/tendon_routing_analysis.py
Tendon pivot	results/tendon_pivot/	scripts/tendon_pivot_optimizer.py
Type B round-1 FEA	TYPE B/analysis/results/	TYPE B/analysis/fea_typeb.py
Type B round-1 report	TYPE B/analysis/REPORT.md	TYPE B/analysis/analyse_results.py
Type B round-2 FEA (new)	TYPE B/round2/fea/results/	TYPE B/round2/fea_round2.py
Type B round-2 test plan	TYPE B/round2/TEST_PLAN.md	TYPE B/round2/predict_round2.py
Material basis (new)	results/sunlu_rescale/	scripts/sunlu_rescale.py